Data Analysis Last Topic:

Coefficient of Determination (R^2)

Definition A statistical measure that indicates the proportion of variance in the dependent variable that is explained by the independent variable(s) in a regression model. It shows how well the model fits the data, ranging from **0 to 1** (or 0% to 100%).

$$R^2 = \frac{a\sum Y - b\sum XY - \frac{(\sum Y)^2}{n}}{\sum Y^2 - \frac{(\sum Y)^2}{n}}$$

Uses

1. Model Evaluation R^2 tells how well a regression model explains the data. For example, R^2 = 0.85 means 85% of the variation in the dependenst variable is explained by the model — the higher the value, the better the fit.

2. Comparing Regression Models When multiple models are tested on the same dataset, R^2 helps select the best one. The model with the highest R^2 (closest to 1) is generally considered the most accurate predictor.

Example:

Advertisement (X)	Sales (Y)
2	5
4	10
6	13
8	17
10	20

Solution:

Given / Calculated Values:

n = 5 | $\Sigma X = 30$ | $\Sigma Y = 65$ | $\Sigma XY = 464$ | $\Sigma X^2 = 220$ | $\Sigma Y^2 = 983$ | a = 1.9 | b = 1.85

$$R^2 = \frac{a\sum Y + b\sum XY - \frac{(\sum Y)^2}{n}}{\sum Y^2 - \frac{(\sum Y)^2}{n}}$$

$$R^2 = \frac{1.9(65) + 1.85(464) - \frac{(65)^2}{5}}{983 - \frac{(65)^2}{5}}$$

$$R^2 = \mathbf{0.992}$$

Interpretation: R^2 = 0.992 means **99.2% of the variation in Sales is explained by Advertisement**, indicating an excellent fit with only 0.8% variation due to other factors.

Assumption of regression Model:

#	Assumption	Description
1	Linearity	Relationship between X and Y must be linear (Scatter plot should be linear)
2	Independence	Observations must be independent of each other (no geometric series) X: 5, 15,25,35,------ ➔ dependent
3	Homoscedasticity	Constant variance of error terms across all levels of X (each point have same variation) Observed – predicted= E➔ Var(E)= σ^2
4	Normality of Errors	Residuals (errors) should be normally distributed (bell shape)
5	No Multicollinearity	*(Multiple regression only)* Independent variables should not be highly correlated with each other (students consumption depends on parent's income & student's pocket money, Dependent variable: Students Consumptions; Independent are parent's Income & student's pocket money where as students pocket money is also dependent on parent's income)
6	No Autocorrelation	Error terms should not be correlated with each other (important in time series) E= 15,20,25,30,------

Time Series:

A time series is a set of statistical observations arranged in **chronological order** at regular intervals of time (hourly, daily, monthly, yearly) to analyze past behavior and forecast future trends.

Components of Time Series:

1. Secular Trend (T) The long-term smooth movement in data over a long period of time. It shows the general direction — upward, downward, or stable. Example: population growth increasing over decades.

2. Seasonal Variations (S) Regular and periodic fluctuations that repeat within a period of **one year or less** due to seasons, customs, or habits. Example: sales of umbrellas rise every monsoon season.

3. Cyclical Variations (C) Long-term wave-like fluctuations that repeat over a period of **more than one year**, usually linked to business cycles of boom and recession. Example: economic expansion followed by depression every few years.

4. Irregular / Random Variations (I) Unpredictable, erratic fluctuations caused by **unforeseen events** such as earthquakes, floods, wars, or strikes. No fixed pattern exists. Example: stock market crash due to a pandemic.

Methods of Identifying Trend Component

1. Freehand (Graphic) Method

A curve or line is drawn **freehand** through the plotted data points on a graph by visual judgment. It is the simplest method but highly subjective as it depends on the drawer's personal judgment. It gives a rough idea of the trend direction.

2. Semi-Average Method

The data is divided into **two equal halves** and the average of each half is calculated. These two averages are plotted and joined by a straight line to represent the trend. It is simple and objective but ignores minor fluctuations.

3. Moving Average Method

Successive **arithmetic averages** are calculated for overlapping groups of a fixed number of periods (3-year, 5-year moving average). Each average replaces the middle value of the group, smoothing out short-term fluctuations and revealing the long-term trend. Most widely used method.

Methods of Identifying Trend — Examples

1. Freehand Method

Year	Sales (Y)
2018	20
2019	24
2020	22
2021	27
2022	30
2023	28
2024	33

Method: Plot the data on graph paper and draw a smooth freehand curve through the points by visual judgment. The rising direction of the curve shows an **upward trend** in sales.

---

2. Semi-Average Method

(A) Odd Number of Years (7 Years)

Year	Sales (Y)
2018	10
2019	12
2020	14
2021	16
2022	18
2023	20
2024	22

Step 1: Divide into two halves (middle year 2021 is excluded)

First Half (2018–2020): 10 + 12 + 14 = 36

Second Half (2022–2024): 18 + 20 + 22 = 60

Year	Sales	Trend
2018	10	—
2019	12	**12**
2020	14	—
2021	16	*(excluded)*
2022	18	—
2023	20	**20**
2024	22	—

Interpretation: Trend rises from 12 to 20 showing a consistent **upward trend** in sales.

(B) Even Number of Years (6 Years)

Year	Sales (Y)
2018	10
2019	14
2020	16
2021	20
2022	22
2023	26

Step 1: Divide into two equal halves of 3 years each

First Half (2018–2020): 10 + 14 + 16 = 40

Second Half (2021–2023): 20 + 22 + 26 = 68

Year	Sales	Trend
2018	10	—
2019	14	**13.3**
2020	16	—
2021	20	—
2022	22	**22.7**
2023	26	—

Interpretation: Trend rises from 13.3 to 22.7 showing a strong **upward trend** in sales.

---

3. Moving Average Method

(A) 3-Year Moving Average

Year	Sales (Y)
2018	10
2019	14
2020	12
2021	18
2022	16
2023	22
2024	20

Year	Sales	3-Year Moving Total	3-Year Moving Average (Trend)
2018	10	—	—
2019	14	10+14+12 = 36	36/3 = **12.0**

Year	Sales	3-Year Moving Total	3-Year Moving Average (Trend)
2020	12	14+12+18 = 44	44/3 = **14.7**
2021	18	12+18+16 = 46	46/3 = **15.3**
2022	16	18+16+22 = 56	56/3 = **18.7**
2023	22	16+22+20 = 58	58/3 = **19.3**
2024	20	—	—

Interpretation: Moving averages rise from 12.0 to 19.3 confirming a clear **upward trend** in sales. First and last years have no trend values.

(B) 4-Year Moving Average (Even)

Year	Sales (Y)
2018	10
2019	14
2020	12
2021	18
2022	16
2023	22
2024	20
2025	24

For even moving averages, **centering** is required:

Year	Sales	4-Year Moving Total	4-Year Moving Avg	Centered (Trend)
2018	10	—	—	—
2019	14	—	—	—

Year	Sales	4-Year Moving Total	4-Year Moving Avg	Centered (Trend)
2020	12	10+14+12+18=54	54/4=13.5	(13.5+15.0)/2=**14.25**
2021	18	14+12+18+16=60	60/4=15.0	(15.0+17.0)/2=**16.00**
2022	16	12+18+16+22=68	68/4=17.0	(17.0+18.5)/2=**17.75**
2023	22	18+16+22+20=76	76/4=19.0	(19.0+19.0)/2=**19.00**
2024	20	16+22+20+24=82	82/4=20.5	—
2025	24	—	—	—

Interpretation: Centered moving averages rise from 14.25 to 19.00 confirming a steady **upward trend** in sales. Centering is necessary in even-period moving averages to align trend values with actual time periods.